# EECE5644 Fall 2026 — Assignment 1

David Abu

October 2026

**Declaration of external resources and AI use.** This submission was prepared with the assistance of an AI coding agent, Claude (Anthropic, model Claude Opus 5.5), used through the claude.ai app. Claude wrote the Python code in the appendix, ran the experiments, and drafted the derivations and discussion; I reviewed, verified and take responsibility for the content. Software used: Python 3, NumPy, SciPy, pandas, Matplotlib. Data: the UCI Wine Quality (white wine) and UCI HAR datasets. Because the UCI website was not reachable from the compute environment, byte-identical copies were obtained from public GitHub mirrors (`github.com/zygmuntz/wine-quality`, `github.com/srvds/Human-Activity-Recognition`); the sample counts (4898 and $7352 + 2947 = 10299$) and feature counts (11 and 561) match the UCI descriptions. No material was exchanged with classmates.

**Notation and loss convention.** $\lambda_{ij}$ is the loss for deciding $D = i$ when the true label is $L = j$. All random data are generated once with fixed seeds, saved to disk, and the same data are used for every part of a question.

## Question 1

Generated $N = 10{,}000$ samples ($L = 1$ with probability 0.65): $N_0 = 3460$, $N_1 = 6540$. Seed 5644; data saved in `q1_data.npz`.

### Part A: ERM with the true pdf

**1. Decision rule.** The conditional risk of decision $i$ is $R(D{=}i \mid \mathbf{x}) = \sum_j \lambda_{ij} P(L{=}j \mid \mathbf{x})$. Deciding $D = 1$ is optimal iff $R(D{=}1|\mathbf{x}) < R(D{=}0|\mathbf{x})$:

$$\begin{aligned} &\lambda_{10} P(L{=}0|\mathbf{x}) + \lambda_{11} P(L{=}1|\mathbf{x}) < \lambda_{00} P(L{=}0|\mathbf{x}) + \lambda_{01} P(L{=}1|\mathbf{x}) \\ \iff\ &(\lambda_{01} - \lambda_{11})\, p(\mathbf{x}|L{=}1) P(L{=}1) > (\lambda_{10} - \lambda_{00})\, p(\mathbf{x}|L{=}0) P(L{=}0), \end{aligned}$$

using Bayes' rule and cancelling $p(\mathbf{x})$. Assuming $\lambda_{01} > \lambda_{11}$ (an error costs more than a correct decision), the minimum-expected-risk rule is the likelihood-ratio test

$$\boxed{\frac{p(\mathbf{x} \mid L=1)}{p(\mathbf{x} \mid L=0)} \underset{D=0}{\overset{D=1}{\gtrless}} \gamma = \frac{(\lambda_{10} - \lambda_{00})\, P(L=0)}{(\lambda_{01} - \lambda_{11})\, P(L=1)}}$$

For minimum probability of error (0–1 loss: $\lambda_{00} = \lambda_{11} = 0$, $\lambda_{01} = \lambda_{10} = 1$): $\gamma^* = P(L{=}0)/P(L{=}1) = 0.35/0.65 = 0.5385$, $\ln \gamma^* = -0.619$. With Gaussian class conditionals the test is implemented in the log domain (numerically stable, monotone so equivalent):

$$\ln \frac{p(\mathbf{x}|1)}{p(\mathbf{x}|0)} = -\tfrac{1}{2}(\mathbf{x} - \mathbf{m}_1)^T C_1^{-1} (\mathbf{x} - \mathbf{m}_1) + \tfrac{1}{2}(\mathbf{x} - \mathbf{m}_0)^T C_0^{-1} (\mathbf{x} - \mathbf{m}_0) - \tfrac{1}{2} \ln \frac{|C_1|}{|C_0|} \ \gtrless\ \ln \gamma.$$

This is a quadratic discriminant, since $C_0 \neq C_1$.

**2. ROC.** The score $s(\mathbf{x}_n) = \ln[p(\mathbf{x}_n|1)/p(\mathbf{x}_n|0)]$ was computed for all samples. Sweeping $\ln \gamma$ from $-\infty$ through every observed score value (10,001 thresholds) gives every distinct operating point the data can produce; for each, $P(D{=}1|L{=}1;\gamma)$ and $P(D{=}1|L{=}0;\gamma)$ are estimated by counts. At $\gamma = 0$ the curve is at $(1, 1)$; at $\ln \gamma = \max_n s(\mathbf{x}_n)$ (finite) it reaches $(0, 0)$. Both $P(D{=}1|L{=}0;\gamma)$ and $P(D{=}0|L{=}1;\gamma) = 1 - P(D{=}1|L{=}1;\gamma)$ are stored for each threshold.

**3. Minimum-$P$(error) threshold.** For each threshold,

$$P(\text{error};\gamma) = P(D{=}1|L{=}0;\gamma)P(L{=}0) + P(D{=}0|L{=}1;\gamma)P(L{=}1);$$

the minimizer is marked on the ROC (Fig. 1). Any $\gamma$ between two consecutive sorted scores gives the same decisions; I report the midpoint. Results are in Table 1.

| Classifier | threshold at min $P(e)$ | $P(D{=}1\|L{=}1)$ | $P(D{=}1\|L{=}0)$ | $\hat{P}$(error) | AUC |
|---|---|---|---|---|---|
| A: true pdf | $\hat{\gamma} = 0.791$ ($\ln\hat{\gamma} = -0.235$) | 0.9679 | 0.0145 | 0.0259 | 0.9958 |
| at theory $\gamma^* = 0.5385$ | $\ln\gamma^* = -0.619$ | 0.9716 | 0.0225 | 0.0264 | |
| B: naive Bayes | $\hat{\gamma} = 0.640$ ($\ln\hat{\gamma} = -0.447$) | 0.9654 | 0.0286 | 0.0325 | 0.9941 |
| at $\gamma^* = 0.5385$ | $\ln\gamma^* = -0.619$ | 0.9670 | 0.0318 | 0.0326 | |
| C: Fisher LDA | $\hat{\tau} = -0.0053$ | 0.9512 | 0.0584 | 0.0521 | 0.9853 |

Table 1: Question 1 summary. Probabilities are sample-count estimates on the same 10K samples.

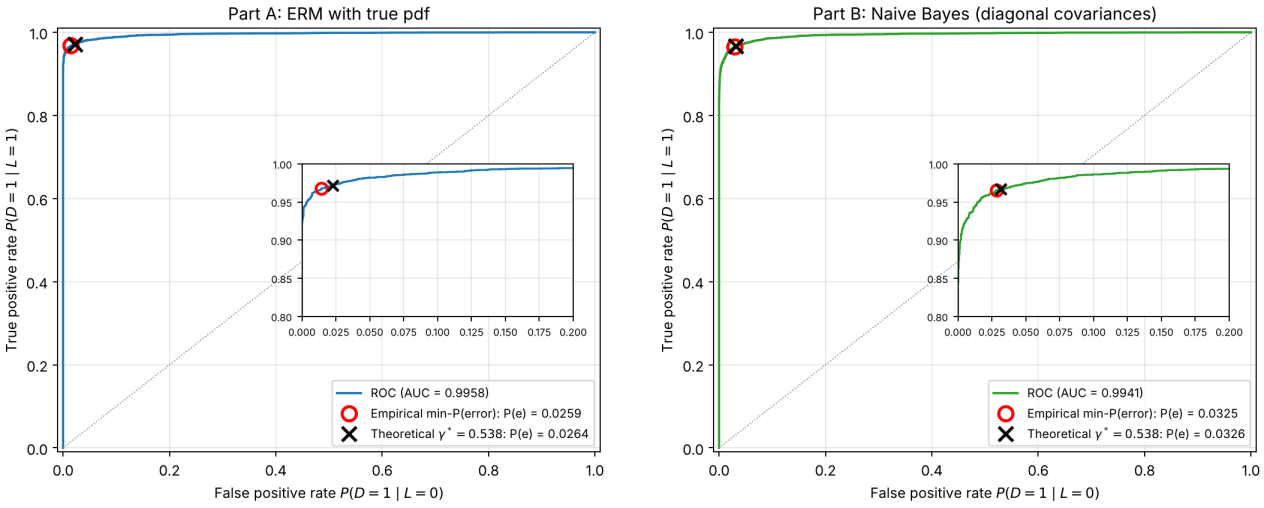



Figure 1: ROC curves of the ERM classifier with the true pdf (left) and the naive Bayes classifier (right). Red circle: empirically selected minimum-$P$(error) operating point; black cross: operating point at the theoretical threshold $\gamma^* = 0.5385$. Insets zoom into the upper-left corner.

**Estimated minimum $P$(error) $\approx$ 0.0259** (at $\hat{\gamma} \approx 0.79$), versus 0.0264 at the theoretical $\gamma^* = 0.538$. As an independent reference, I evaluated the theoretical rule on a separate 2-million-sample Monte Carlo set: $P(\text{error};\gamma^*) \approx 0.0262$, which is the Bayes error of this distribution to about $\pm 0.0001$.

**Comparison of empirical vs. theoretical $\gamma$.** The empirical minimizer ($\hat{\gamma} \in [0.783, 0.799]$) is larger than $\gamma^* = 0.538$, yet the two error values differ by only 0.00045 (about 4–5 of 10,000 samples). Figure 2 shows why: $P(\text{error};\gamma)$ is very flat around its minimum, so with a finite sample the location of the minimum is noisy. The empirical $\hat{\gamma}$ is fitted to these particular 10K samples, so its error 0.0259 is optimistically biased (below the large-sample Bayes error 0.0262). The theoretically derived $\gamma^*$ is the one that is optimal for the distribution; the empirical one only agrees with it up to sampling noise, and would converge to it as $N \to \infty$.

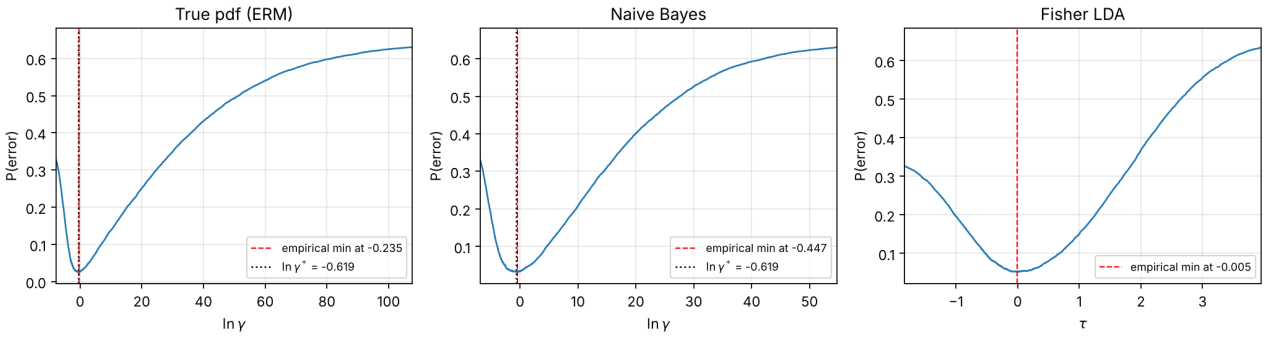



Figure 2: Estimated $P(\text{error})$ versus threshold for the three classifiers. Red dashed: empirical minimizer; black dotted: $\ln \gamma^*$.

## Part B: Naive Bayes (diagonal covariance) model mismatch

The classifier uses the same LRT and threshold formula but with $C_0^{NB} = \text{diag}(C_0) = \text{diag}(0.5, 0.25, 0.25, 0.5)$ and $C_1^{NB} = \text{diag}(C_1) = \text{diag}(1, 2, 1, 3)$, i.e. it ignores the correlations among $x_1, x_2, x_3$ ($x_4$ is already uncorrelated). The score is $\ln g(\mathbf{x}|\mathbf{m}_1, C_1^{NB}) - \ln g(\mathbf{x}|\mathbf{m}_0, C_0^{NB})$ and the same sweep is performed on the same 10K samples (Fig. 1 right, Table 1).

- Minimum $\hat{P}(\text{error}) = \mathbf{0.0325}$ at $\hat{\gamma} \approx 0.640$ (vs. 0.0326 at $\gamma^* = 0.538$). Monte Carlo with 2M samples at $\gamma^*$: 0.0322.
- The ROC moves down/right (AUC $0.9958 \to 0.9941$) and the best achievable error rises by about 25% relative ($0.0259 \to 0.0325$; $0.0262 \to 0.0322$ with 2M samples).
- Here $\gamma^*$ from priors and losses is no longer guaranteed optimal, because the NB ratio is not the true likelihood ratio, so thresholding it at $\gamma^*$ is not the Bayes rule. Empirically the best threshold (0.64) is still close to $\gamma^*$ and the error at $\gamma^*$ is almost the same as the minimum.

**Conclusion:** yes, the mismatch hurts, but moderately. The decision boundary is still quadratic and uses the correct means and variances; it only has the wrong orientation of the quadrics in the $(x_1, x_2, x_3)$ subspace. The correlations ($|\rho| \leq 0.5$) are moderate and the class means are well separated (distance 3), so the loss is noticeable but not dramatic.

## Part C: Fisher LDA

Sample-average estimates from the 10K samples:

$$\hat{\mathbf{m}}_0 = (-0.500, -0.505, -0.496, -0.498)^T, \quad \hat{\mathbf{m}}_1 = (1.011, 1.016, 0.995, 0.983)^T,$$

and

$$\hat{C}_j = \tfrac{1}{N_j} \textstyle\sum_{n:L_n=j} (\mathbf{x}_n - \hat{\mathbf{m}}_j)(\mathbf{x}_n - \hat{\mathbf{m}}_j)^T$$

(both close to the true $C_0, C_1$). With equal weights, $S_B = (\hat{\mathbf{m}}_1 - \hat{\mathbf{m}}_0)(\hat{\mathbf{m}}_1 - \hat{\mathbf{m}}_0)^T$ and $S_W = \hat{C}_0 + \hat{C}_1$. The Fisher criterion $J(\mathbf{w}) = \frac{\mathbf{w}^T S_B \mathbf{w}}{\mathbf{w}^T S_W \mathbf{w}}$ is maximized by the generalized eigenvector $S_B\mathbf{w} = \lambda S_W \mathbf{w}$ with the largest eigenvalue ($\lambda_{\max} = 4.689$; $S_B$ has rank 1 so this is the only nonzero one). Normalized and oriented so that class 1 projects higher,

$$\mathbf{w}_{LDA} = (0.6028,\ 0.2747,\ 0.7086,\ 0.2432)^T,$$

which equals the closed form $S_W^{-1}(\hat{\mathbf{m}}_1 - \hat{\mathbf{m}}_0)/\|\cdot\|$ (checked numerically). The rule is $D = 1$ iff $\mathbf{w}_{LDA}^T \mathbf{x} > \tau$; sweeping $\tau$ over $(-\infty, \infty)$ (all projected values) gives the ROC in Fig. 3. The error at each $\tau$ is estimated by sample counts, $\#\{\text{errors}\}/N$.

Minimum $\hat{P}(\text{error}) = \mathbf{0.0521}$ at $\hat{\tau} \approx -0.0053$ ($P(D{=}1|L{=}1) = 0.951$, $P(D{=}1|L{=}0) = 0.058$), AUC = 0.9853.

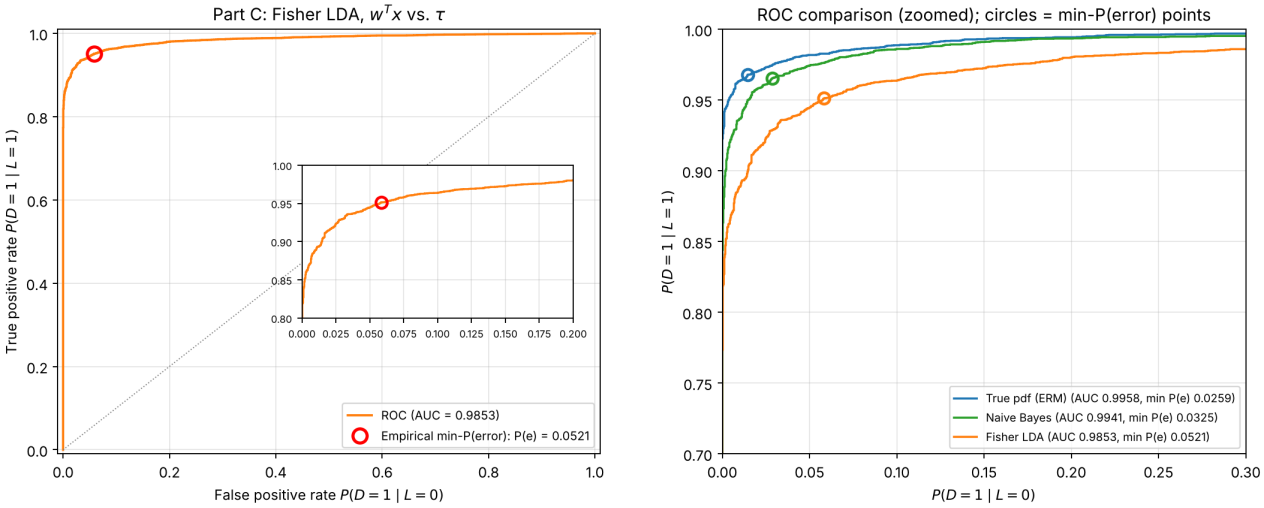



Figure 3: Left: Fisher LDA ROC with the minimum-$P$(error) point. Right: zoomed comparison of the three ROC curves.

**Discussion.** The ranking is true-pdf ERM (0.0259) < naive Bayes (0.0325) < LDA (0.0521), and the ROC curves are nested the same way (Fig. 3 right). LDA roughly doubles the error of the optimal classifier. The reason is structural: the Bayes rule is quadratic because $C_1$ has much larger variances than $C_0$ (e.g. var($x_4$) is 3 vs. 0.5), so the optimal boundary is a closed-ish surface wrapped around the tight class-0 cloud. LDA can only use a hyperplane, and Fisher's criterion (one pooled within-class scatter, equal weights) is only optimal for equal-covariance Gaussians. Naive Bayes, although its covariances are wrong, still exploits the covariance difference through a quadratic boundary, which is why it beats LDA here. LDA also ignores the priors in finding $\mathbf{w}$, but this does not matter for the result since the threshold $\tau$ is selected empirically. LDA's advantages are that it needs no distributional knowledge and only $d$ projection parameters.

# Question 2

## Data distribution

Four 3-D Gaussian components with means at the vertices of a regular tetrahedron of edge $a = 2.5$, so every pair of means is 2.5 apart:

$$\boldsymbol{\mu}_1 = (0,0,0),\ \boldsymbol{\mu}_2 = (2.5,0,0),\ \boldsymbol{\mu}_3 = (1.25,2.165,0),\ \boldsymbol{\mu}_4 = (1.25,0.722,2.041),$$

$$\Sigma_1 = \begin{bmatrix} 1 & .3 & -.2 \\ .3 & .8 & .1 \\ -.2 & .1 & 1.2 \end{bmatrix}, \Sigma_2 = \begin{bmatrix} .9 & -.2 & .1 \\ -.2 & 1.1 & .3 \\ .1 & .3 & .8 \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 1.2 & .2 & 0 \\ .2 & .9 & -.3 \\ 0 & -.3 & 1 \end{bmatrix}, \Sigma_4 = \begin{bmatrix} .8 & 0 & .2 \\ 0 & 1 & 0 \\ .2 & 0 & 1.1 \end{bmatrix}.$$

The average standard deviation is $\frac{1}{4}\sum_k \sqrt{\mathrm{tr}(\Sigma_k)/3} = 0.991$, so mean separations are $\approx 2.5\times$ the average standard deviation, giving substantial overlap, as requested. Class conditionals: $p(\mathbf{x}|L{=}1) = g(\mathbf{x}|\boldsymbol{\mu}_1, \Sigma_1)$, $p(\mathbf{x}|L{=}2) = g(\mathbf{x}|\boldsymbol{\mu}_2, \Sigma_2)$, $p(\mathbf{x}|L{=}3) = \frac{1}{2}g(\mathbf{x}|\boldsymbol{\mu}_3, \Sigma_3) + \frac{1}{2}g(\mathbf{x}|\boldsymbol{\mu}_4, \Sigma_4)$. Priors $(0.3, 0.3, 0.4)$.

## Part A: MAP classification

**1.** 10,000 samples were generated (seed 2026) by drawing $L$, then the component (fair coin for class 3), then $\mathbf{x}$. Counts: $N_1 = 2950$, $N_2 = 2993$, $N_3 = 4057$.

**2. Decision rule.** With 0–1 loss, $R(D{=}i|\mathbf{x}) = 1 - P(L{=}i|\mathbf{x})$, so the minimum-$P$(error) (MAP) rule is

$$D(\mathbf{x}) = \arg\max_{j\in\{1,2,3\}} P(L{=}j|\mathbf{x}) = \arg\max_j \left[\ln p(\mathbf{x}|L{=}j) + \ln P(L{=}j)\right],$$

with $\ln p(\mathbf{x}|L{=}3)$ computed via log-sum-exp of the two components. The estimated confusion matrix, with entries $\hat{P}(D{=}i|L{=}j) = \#\{D{=}i, L{=}j\}/N_j$, is (counts in parentheses; columns sum to 1):

| | $L=1$ | $L=2$ | $L=3$ |
|---|---|---|---|
| $D=1$ | **0.7963** (2349) | 0.0672 (201) | 0.0845 (343) |
| $D=2$ | 0.0600 (177) | **0.8199** (2454) | 0.0656 (266) |
| $D=3$ | 0.1437 (424) | 0.1129 (338) | **0.8499** (3448) |

Estimated $P(\text{error}) = \mathbf{0.1749}$. This is also the estimated minimum expected risk under 0–1 loss.

**3. Visualization.** Fig. 4: marker shape encodes the true label (dot $L$=1, triangle $L$=2, square $L$=3); green means correct, red incorrect. Errors concentrate in the central region where all four components overlap.

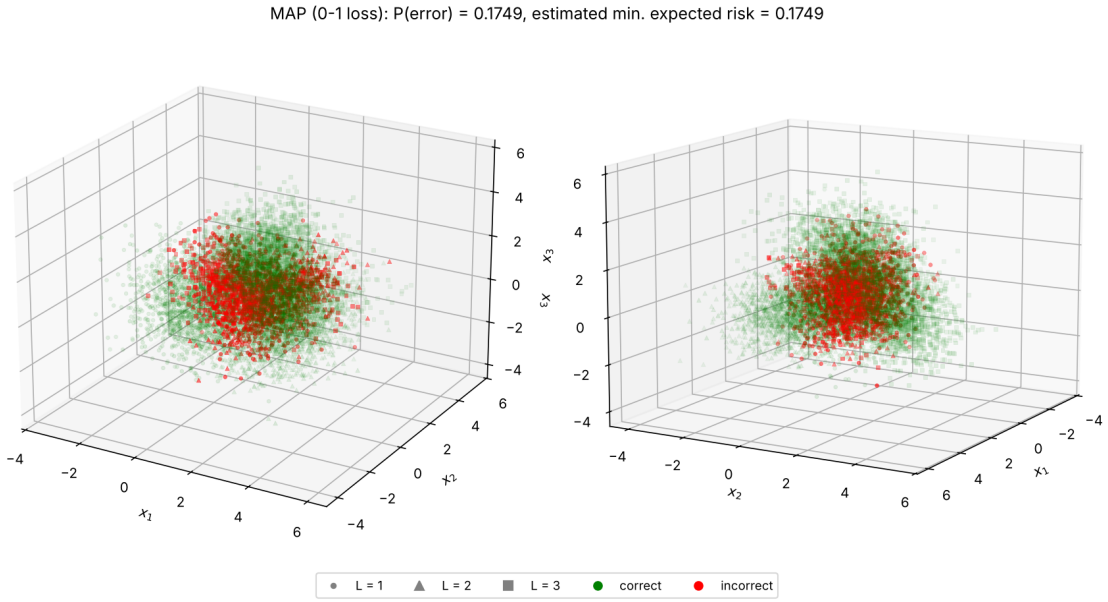



Figure 4: MAP classification of the 10K samples, two viewing angles.

## Part B: ERM with $\Lambda_{10}$ and $\Lambda_{100}$

The ERM rule is $D(\mathbf{x}) = \arg\min_i \sum_j \lambda_{ij} P(L{=}j|\mathbf{x})$. For $\Lambda_w = \begin{bmatrix} 0 & 1 & w \\ 1 & 0 & w \\ 1 & 1 & 0 \end{bmatrix}$ (with $P_j \equiv P(L{=}j|\mathbf{x})$):

$$R(1|\mathbf{x}) = P_2 + wP_3, \quad R(2|\mathbf{x}) = P_1 + wP_3, \quad R(3|\mathbf{x}) = P_1 + P_2 = 1 - P_3.$$

Between 1 and 2 the $wP_3$ term cancels, so that choice is still by larger posterior. Class 3 is chosen iff $1 - P_3 < \min(P_1, P_2) + wP_3$, i.e.

$$P(L{=}3|\mathbf{x}) > \frac{1 - \min\{P(L{=}1|\mathbf{x}), P(L{=}2|\mathbf{x})\}}{w+1}.$$

The right side lies between $\frac{1}{2(w+1)}$ and $\frac{1}{w+1}$. So with $w = 10$, class 3 is chosen once its posterior exceeds at most $1/11 \approx 0.09$; with $w = 100$, once it exceeds at most $1/101 \approx 0.01$.

The minimum expected risk is estimated by the sample average of the loss actually incurred, $\hat{R}_{\min} = \frac{1}{N}\sum_n \lambda_{D(\mathbf{x}_n),L_n}$. A second, lower-variance estimate is $\frac{1}{N}\sum_n \min_i R(D{=}i|\mathbf{x}_n)$, which uses the true posteriors. Both agree closely:

| Loss | $\hat{R}_{\min} = \frac{1}{N}\sum_n \lambda_{D_n L_n}$ | $\frac{1}{N}\sum_n \min_i R(D{=}i\|x_n)$ | $\hat{P}(\text{error})$ | $\#\{D=1,2,3\}$ |
|---|---|---|---|---|
| 0–1 | 0.1749 | 0.1775 | 0.1749 | 2893, 2897, 4210 |
| $\Lambda_{10}$ | 0.3387 | 0.3451 | 0.2757 | 1567, 1933, 6500 |
| $\Lambda_{100}$ | 0.4882 | 0.4899 | 0.4387 | 550, 1063, 8387 |

Confusion matrices $\hat{P}(D{=}i|L{=}j)$ for $\Lambda_{10}$ (left) and $\Lambda_{100}$ (right):

| | $L=1$ | $L=2$ | $L=3$ |
|---|---|---|---|
| $D=1$ | **0.4912** (1449) | 0.0267 (80) | 0.0094 (38) |
| $D=2$ | 0.0319 (94) | **0.6037** (1807) | 0.0079 (32) |
| $D=3$ | 0.4769 (1407) | 0.3695 (1106) | **0.9827** (3987) |

| | $L=1$ | $L=2$ | $L=3$ |
|---|---|---|---|
| $D=1$ | **0.1797** (530) | 0.0060 (18) | 0.0005 (2) |
| $D=2$ | 0.0098 (29) | **0.3445** (1031) | 0.0007 (3) |
| $D=3$ | 0.8105 (2391) | 0.6495 (1944) | **0.9988** (4052) |

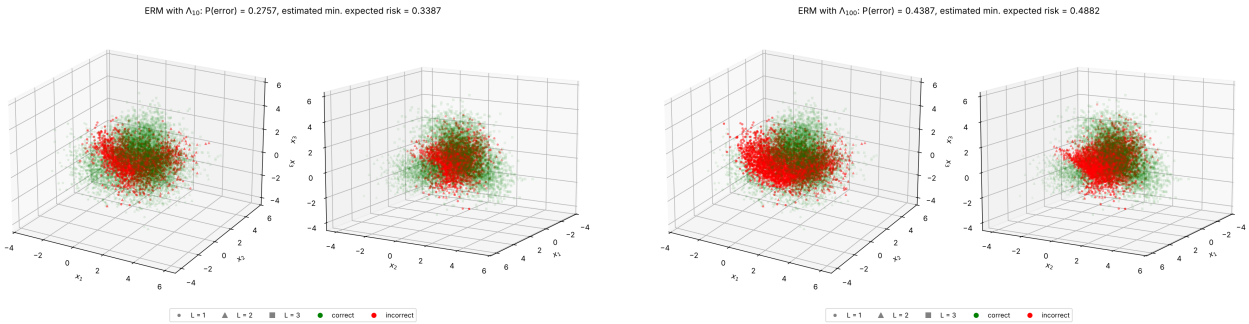



Figure 5: ERM decisions with $\Lambda_{10}$ (left) and $\Lambda_{100}$ (right); same marker/colour convention as Fig. 4.

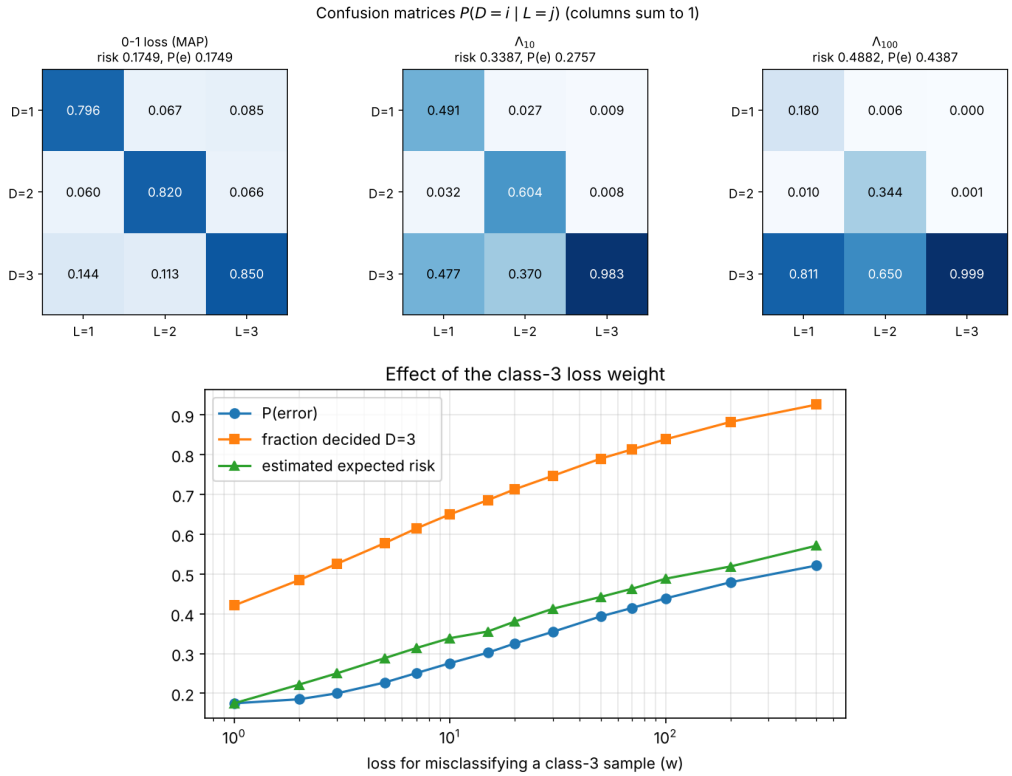



Figure 6: Top: confusion matrices for the three loss matrices. Bottom: P(error), fraction of $D=3$ decisions, and estimated expected risk as the class-3 loss weight $w$ varies from 1 (0–1 loss) to 500.

**Insights.**

- *Risk vs. error.* As $w$ grows, ERM trades many cheap errors for few expensive ones. Class-3 recall goes $0.850 \to 0.983 \to 0.999$, while recall of classes 1 and 2 collapses ($0.796 \to 0.491 \to 0.180$ and $0.820 \to 0.604 \to 0.345$). The number of $D = 3$ decisions rises from 4210 to 6500 to 8387. $P(\text{error})$ rises sharply ($0.175 \to 0.276 \to 0.439$), but this is the risk-optimal behaviour, not a failure.
- *Minimum risk grows with $w$* ($0.175 \to 0.339 \to 0.488$): larger penalties cannot reduce the best achievable risk. However, the growth is strongly sub-linear in $w$ because the classifier adapts. Under $\Lambda_{100}$ only 5 of 4057 class-3 samples are missed, yet those 5 still contribute $500/10^4 = 0.05$ to the risk. The remaining $\approx 0.44$ comes from the 4335 class-1/2 samples labelled 3 at cost 1 each.
- *Asymmetry between classes 1 and 2:* class 1 loses more recall than class 2. Because of the covariance orientations, more of class 1's probability mass lies where $P(L{=}3|\mathbf{x})$ exceeds the low class-3 threshold.
- In the limit $w \to \infty$ the rule degenerates to "always decide 3", with risk 0.6 (all class-1/2 samples wrong at cost 1). The sweep shows the risk approaching that value (0.571 at $w = 500$).

# Question 3

## Model, estimators and decision rule

For each dataset with classes $j \in \mathcal{C}$ and samples $\{(\mathbf{x}_n, L_n)\}_{n=1}^N$, $\mathbf{x}_n \in \mathbb{R}^d$:

1. **Assumptions.** Samples are i.i.d.; the class-conditional pdf of each class is a single multivariate Gaussian, $p(\mathbf{x}|L{=}j) = g(\mathbf{x}|\mathbf{m}_j, C_j)$, with its own full covariance (heteroscedastic, i.e. a quadratic-discriminant model). No independence between features is assumed.
2. **Preprocessing.** Each feature is z-scored with the global mean and standard deviation, $\tilde{x}_k = (x_k - \bar{x}_k)/s_k$. An invertible affine map changes every $\ln p(\mathbf{x}|j)$ by the same constant (the log-Jacobian) and leaves an *unregularized* Gaussian classifier's decisions unchanged. Regularization with $\lambda I$ is *not* affine-invariant, though, and wine features differ in variance by $\sim 10^8$ (e.g. total $SO_2$ vs. density). Standardizing makes $\lambda I$ an isotropic shrinkage in comparable units. Without it, the same classifier on raw wine features has training error 0.600 instead of 0.503, because $\lambda$ swamps the small-variance features.
3. **Estimates (maximum likelihood / sample averages).** $\hat{P}(L{=}j) = N_j/N$; $\hat{\mathbf{m}}_j = \frac{1}{N_j}\sum_{n:L_n=j}\mathbf{x}_n$; $\hat{C}_j = \frac{1}{N_j}\sum_{n:L_n=j}(\mathbf{x}_n - \hat{\mathbf{m}}_j)(\mathbf{x}_n - \hat{\mathbf{m}}_j)^T$.
4. **Regularization** (per class, following the hint): $C_j^{reg} = \hat{C}_j + \lambda_j I$ with $\lambda_j = \alpha\ \mathrm{tr}(\hat{C}_j)/\mathrm{rank}(\hat{C}_j)$, the arithmetic mean of the nonzero eigenvalues scaled by $\alpha$. All eigenvalues of $C_j^{reg}$ are then $\geq \lambda_j > 0$.
5. **Decision rule** (0–1 loss, MAP): $D(\mathbf{x}) = \arg\max_j \left[-\frac{1}{2}(\mathbf{x}-\hat{\mathbf{m}}_j)^T(C_j^{reg})^{-1}(\mathbf{x}-\hat{\mathbf{m}}_j) - \frac{1}{2}\ln|C_j^{reg}| + \ln\hat{P}(L{=}j)\right]$, evaluated with Cholesky factors.
6. **Choosing $\alpha$.** Training error on all samples is not a valid criterion: it decreases as $\alpha \to 0$ because the model memorizes. I therefore selected $\alpha$ on held-out data and then, as the assignment requires, re-estimated all parameters on all samples and report the all-sample (training) error and confusion matrix. Wine: stratified 5-fold cross-validation, giving $\alpha = 10^{-3}$ (CV error is flat, about 0.515, for all $\alpha \leq 10^{-2}$). HAR: the dataset's own subject-disjoint train/test split, giving $\alpha = 10^{-4}$. The sweep is in Table 2 and Fig. 7.

| $\alpha$ | Wine: all-sample | Wine: 5-fold CV | HAR: all-sample | HAR: held-out (official split) |
|---|---|---|---|---|
| 1e-06 | 0.5002 | 0.5155 | 0.0217 | 0.0859 |
| 1e-05 | 0.5002 | 0.5155 | 0.0118 | 0.0825 |
| 0.0001 | 0.5008 | 0.5157 | 0.0030 | 0.0818 |
| 0.001 | 0.5033 | 0.5149 | 0.0149 | 0.1042 |
| 0.01 | 0.5043 | 0.5184 | 0.0652 | 0.1259 |
| 0.03 | 0.5065 | 0.5214 | 0.0938 | 0.1367 |
| 0.1 | 0.5159 | 0.5300 | 0.1482 | 0.2009 |
| 0.3 | 0.5208 | 0.5300 | 0.2633 | 0.3258 |

Table 2: P(error) versus regularization strength $\alpha$.

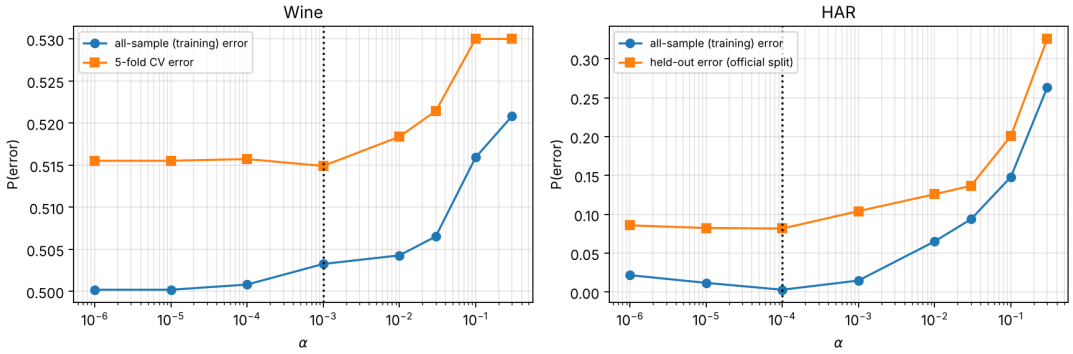



Figure 7: Effect of $\alpha$; dotted line = selected value.

## Results: Wine Quality (white, $N = 4898$, $d = 11$)

Seven quality levels occur (3–9); levels 0–2 and 10 have no samples and are therefore not classes of the model. Per-class estimates:

| Class | $N_j$ | $\hat{P}(L{=}j)$ | rank($\hat{C}_j$) | $\lambda_j$ | cond($\hat{C}_j$) | cond($\hat{C}_j + \lambda_j I$) |
|---|---|---|---|---|---|---|
| 3 | 20 | 0.0041 | 11 | 3.47e-03 | 1.31e+03 | 1.07e+03 |
| 4 | 163 | 0.0333 | 11 | 1.36e-03 | 1.54e+02 | 1.47e+02 |
| 5 | 1457 | 0.2975 | 11 | 9.93e-04 | 2.18e+02 | 2.03e+02 |
| 6 | 2198 | 0.4488 | 11 | 9.37e-04 | 1.11e+02 | 1.07e+02 |
| 7 | 880 | 0.1797 | 11 | 7.71e-04 | 3.06e+02 | 2.83e+02 |
| 8 | 175 | 0.0357 | 11 | 8.86e-04 | 5.42e+02 | 4.74e+02 |
| 9 | 5 | 0.0010 | 4 | 1.28e-03 | 5.89e+17 | 2.54e+03 |

Class 9 has only 5 samples in 11 dimensions, so $\hat{C}_9$ has rank 4 and is singular (condition number $\sim 10^{17}$). Regularization is mandatory here; the other classes are well conditioned.

**Estimated $P$(error) = 0.5033** ($\alpha = 10^{-3}$, all samples). Confusion matrix $\hat{P}(D{=}i|L{=}j)$:

| | $L=3$ | $L=4$ | $L=5$ | $L=6$ | $L=7$ | $L=8$ | $L=9$ |
|---|---|---|---|---|---|---|---|
| $D=3$ | **0.450** | 0.018 | 0.006 | 0.002 | 0.000 | 0.000 | 0.000 |
| $D=4$ | 0.050 | **0.233** | 0.036 | 0.012 | 0.002 | 0.000 | 0.000 |
| $D=5$ | 0.150 | 0.417 | **0.570** | 0.221 | 0.032 | 0.017 | 0.000 |
| $D=6$ | 0.200 | 0.252 | 0.279 | **0.405** | 0.216 | 0.280 | 0.000 |
| $D=7$ | 0.100 | 0.074 | 0.107 | 0.353 | **0.733** | 0.617 | 0.000 |
| $D=8$ | 0.050 | 0.006 | 0.001 | 0.007 | 0.017 | **0.086** | 0.000 |
| $D=9$ | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | **1.000** |

For reference: always guessing the majority class (6) gives error 0.551; predicting within $\pm 1$ grade succeeds for 92.7% of samples.

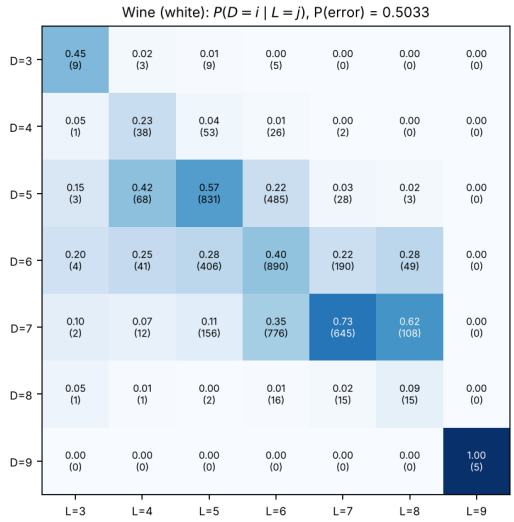



Figure 8: Wine confusion matrix (counts in parentheses).

## Results: Human Activity Recognition ($N = 10299$, $d = 561$)

| Class | $N_j$ | $\hat{P}(L{=}j)$ | rank($\hat{C}_j$) | $\lambda_j$ | cond($\hat{C}_j$) | cond($\hat{C}_j + \lambda_j I$) |
|---|---|---|---|---|---|---|
| 1 Walk | 1722 | 0.1672 | 468 | 7.35e-05 | 1.5e+35 | 1.15e+06 |
| 2 Up | 1544 | 0.1499 | 468 | 6.48e-05 | 1.7e+35 | 9.19e+05 |
| 3 Down | 1406 | 0.1365 | 469 | 1.21e-04 | 3.2e+36 | 1.50e+06 |
| 4 Sit | 1777 | 0.1725 | 468 | 3.80e-05 | 1.0e+36 | 1.15e+06 |
| 5 Stand | 1906 | 0.1851 | 468 | 3.22e-05 | 3.7e+35 | 9.57e+05 |
| 6 Lay | 1944 | 0.1888 | 468 | 4.96e-05 | 5.6e+36 | 1.05e+06 |

Every class covariance has rank only 468–469 even though $N_j \gg d$. Many of the 561 features are exact deterministic functions of others (e.g. magnitudes, energies and band sums derived from the same signals), so the raw $\hat{C}_j$ are numerically singular (condition numbers $\sim 10^{35}$). Regularization brings them to $\sim 10^{6}$.

**Estimated $P(\textbf{error}) = \mathbf{0.0030}$** ($\alpha = 10^{-4}$, all samples, 31 errors). Confusion matrix:

| | $L$=Walk | $L$=Up | $L$=Down | $L$=Sit | $L$=Stand | $L$=Lay |
|---|---|---|---|---|---|---|
| $D$=Walk | **0.9994** | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| $D$=Up | 0.0006 | **1.0000** | 0.0085 | 0.0000 | 0.0000 | 0.0000 |
| $D$=Down | 0.0000 | 0.0000 | **0.9915** | 0.0000 | 0.0000 | 0.0000 |
| $D$=Sit | 0.0000 | 0.0000 | 0.0000 | **0.9983** | 0.0079 | 0.0000 |
| $D$=Stand | 0.0000 | 0.0000 | 0.0000 | 0.0017 | **0.9921** | 0.0000 |
| $D$=Lay | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | **1.0000** |

On held-out subjects (official split: train on 7352 samples, test on 2947) the same model has error 0.082.

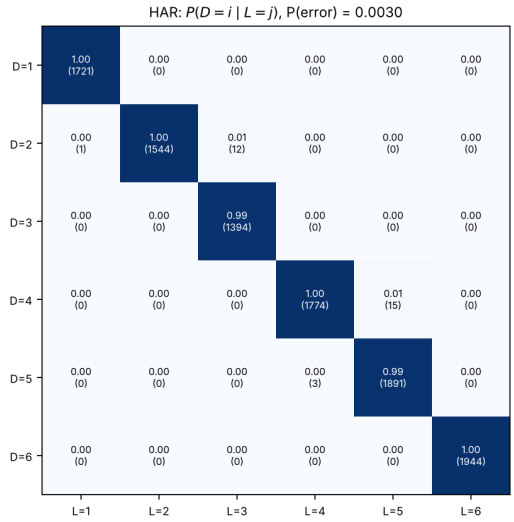



Figure 9: HAR confusion matrix (counts in parentheses).

## Visualizations

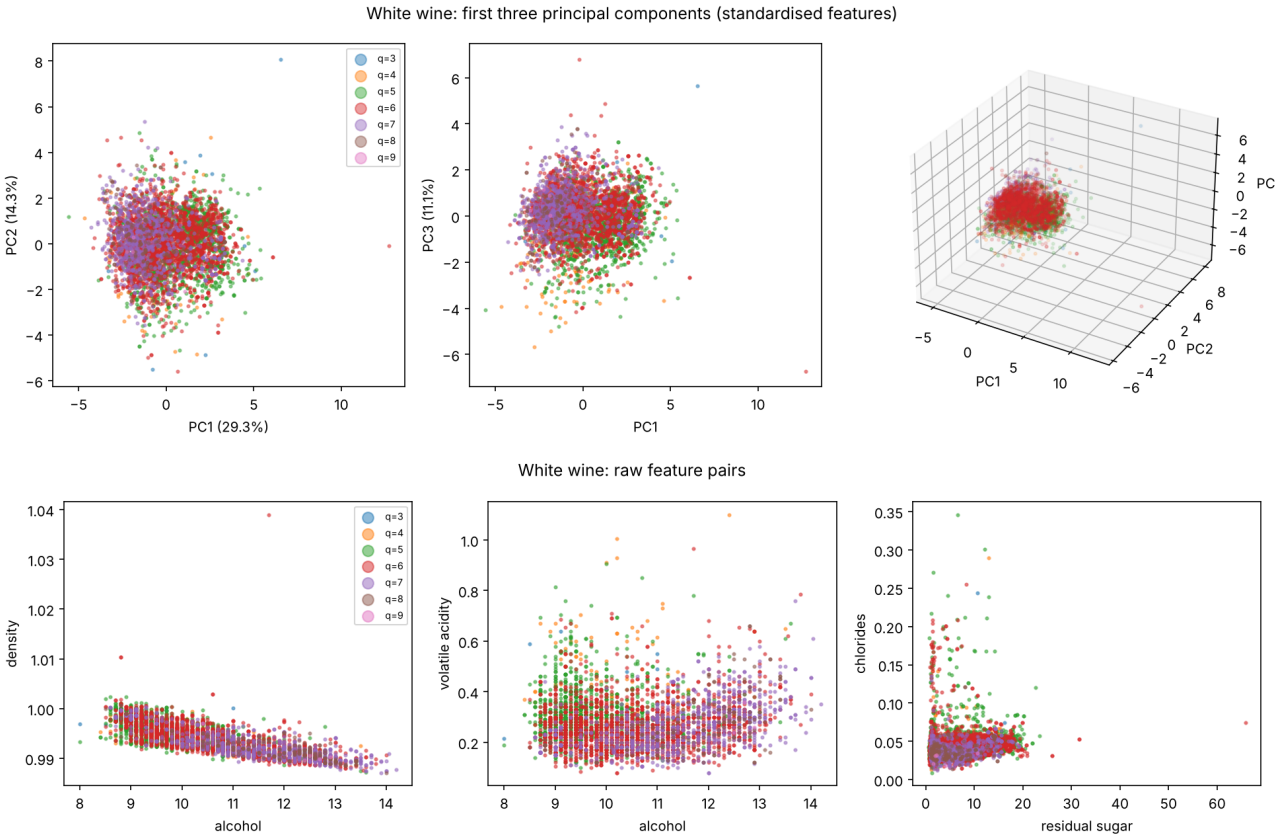



Figure 10: Wine: first three principal components of the standardized features (PC1–3 explain 29%, 14%, 11% of variance), and three raw feature pairs.

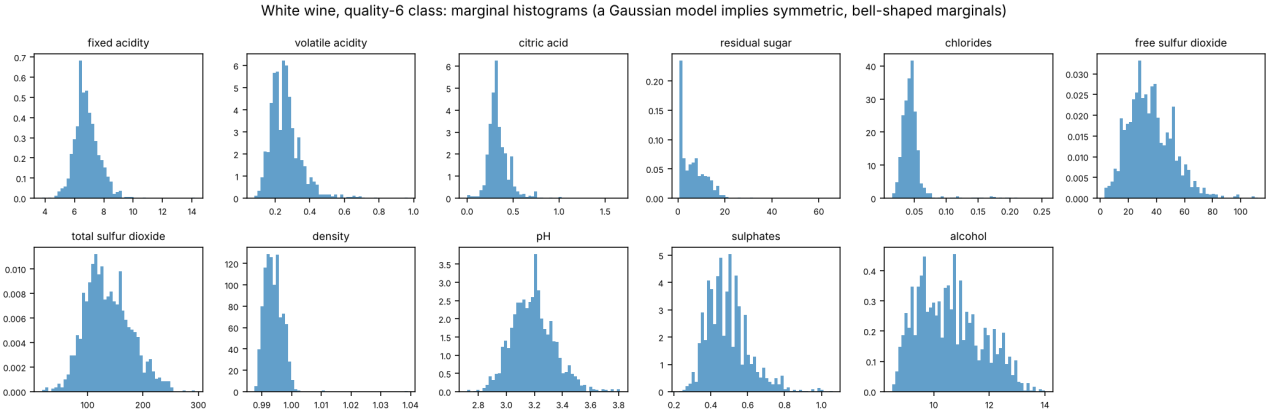



Figure 11: Wine: marginal histograms of all 11 features within the quality-6 class.

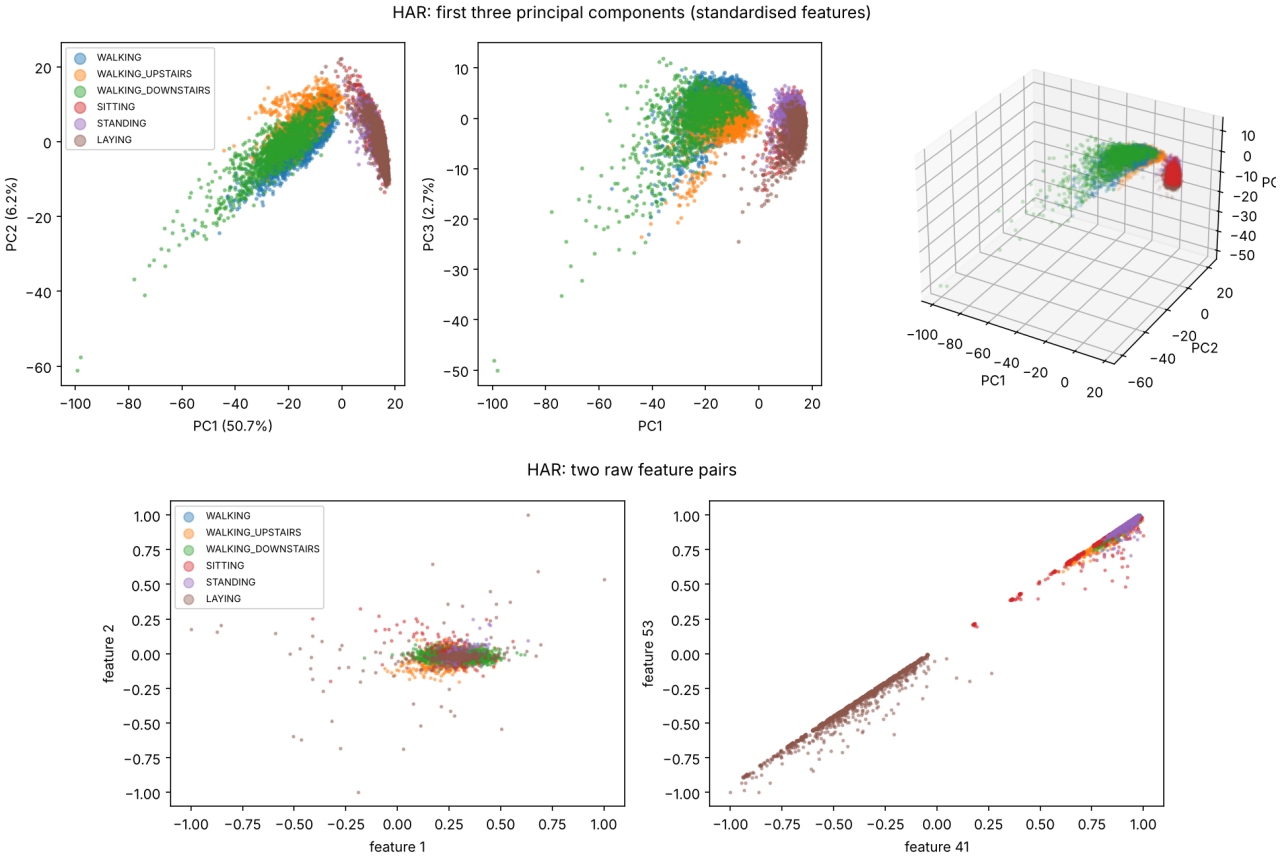



Figure 12: HAR: first three principal components (PC1 alone explains 50.7% of variance) and two raw feature pairs.

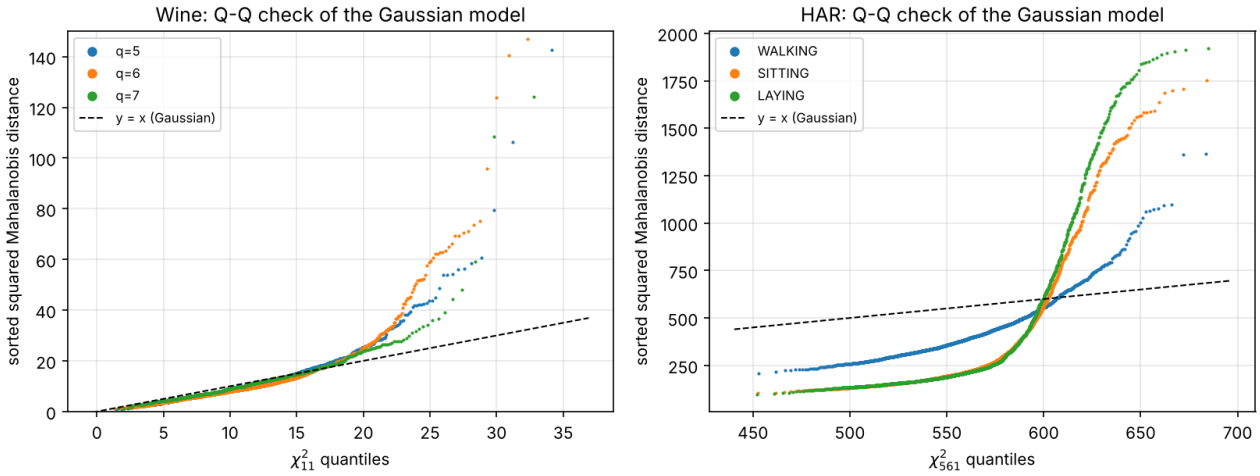



Figure 13: Gaussianity check. If $\mathbf{x} \sim \mathcal{N}(\mathbf{m}_j, C_j)$, the squared Mahalanobis distance $r^2 = (\mathbf{x} - \mathbf{m}_j)^T C_j^{-1} (\mathbf{x} - \mathbf{m}_j) \sim \chi^2_d$. The plot shows sorted $r^2$ (with the fitted, regularized model) against $\chi^2_d$ quantiles; a Gaussian class would lie on the dashed line.

## Discussion: are Gaussian class conditionals appropriate?

**Wine — poorly, and the problem itself is hard.**

- *Non-Gaussian features.* Several marginals are strongly right-skewed or heavy-tailed (residual sugar, chlorides, free $SO_2$, volatile acidity; Fig. 11). The Q-Q plot (Fig. 13, left) follows $\chi^2_{11}$ up to roughly the 80th percentile, then shows much heavier tails: outliers sit at Mahalanobis distances a Gaussian would essentially never produce. A log transform of the skewed features or a mixture/robust model would match better.
- *Overlap.* In every projection (Fig. 10) the quality classes overlap almost completely and differ mostly by small mean shifts along alcohol/density. The labels are subjective, ordinal taster scores, so even the true Bayes error is likely high. The confusion matrix shows this: nearly all errors are to adjacent grades (within $\pm 1$: 92.7%), and the Gaussian model's $P(\text{error}) \approx 0.50$ is only modestly better than the majority-class baseline (0.55).
- *How the model shaped the confusion matrix.* With near-identical means, the quadratic rule is driven largely by covariance differences and priors. Class 6 is the largest class (prior 0.45) but has only 40.5% recall; 22% and 35% of its samples are assigned to the neighbouring grades 5 and 7, whose Gaussians overlap it almost completely. Rare grades are handled very differently. Class 9 (5 samples) gets a tiny, regularized Gaussian that recognizes its own 5 training points (5/5 correct), which is pure memorization. Class 3 (20 samples) gets 45% recall on training data. With 77 parameters per class ($11 + 66$), these rare-class estimates have very high variance, and their training-set performance is not trustworthy. CV error (0.515) slightly above the training error (0.503) confirms mild overfitting overall.

**HAR — not Gaussian, but the classes are so well separated that a Gaussian quadratic classifier still works very well.**

- *Non-Gaussianity.* The features are bounded in $[-1, 1]$ by construction and come from many subjects, so each class is a mixture of per-subject clusters. The Q-Q plot (Fig. 13, right) is far from $\chi^2_{561}$: distances are too small for most samples (the data effectively live in a $\approx$ 468-dimensional subspace, and $\lambda$ inflates the null directions) and far too large in the tail. Strictly, the Gaussian model is wrong.
- *Separation.* PCA (Fig. 12) shows two groups split cleanly along PC1: dynamic activities (walking, upstairs, downstairs) and static ones (sitting, standing, laying). Laying is also distinct. The only confusions are within the static group (sitting↔standing: 18 samples) and within the dynamic group (12 downstairs and 1 walking sample decided as upstairs), exactly the physically similar pairs. In 561 dimensions, a quadratic boundary with class-specific covariances is flexible enough to separate these classes even though each class is not Gaussian.
- *Overfitting and the role of $\lambda$.* Each class model has $561 + 561 \cdot 562/2 \approx 158{,}000$ parameters estimated from $\sim 1{,}700$ samples. The all-sample error of 0.30% is therefore heavily optimistic: on unseen subjects the error is 8.2%. Table 2 shows the bias–variance trade-off. Large $\alpha$ (strong shrinkage toward a spherical covariance)

underfits (error grows to 26–33% at $\alpha = 0.3$); tiny $\alpha$ leaves the near-null eigen-directions, which are dominated by numerical noise, nearly unpenalized. Training error is then not even monotone in $\alpha$, and held-out error is best at intermediate $\alpha = 10^{-4}$ to $10^{-5}$.

**Summary.** The minimum-$P$(error) rule is only optimal if the assumed pdfs are correct. For wine, model mismatch (skewed, heavy-tailed, almost fully overlapping classes) plus severe class imbalance yields $\approx 50\%$ error concentrated between adjacent grades. For HAR, the Gaussian model is also mismatched, but the class-conditional densities are concentrated in different regions, so the induced quadratic boundaries still achieve near-zero training error and $\approx 8\%$ held-out error. In both cases regularization was essential: it made singular covariance estimates invertible (wine class 9; all HAR classes), and its strength set the bias–variance balance.

# Appendix: Code

All code is Python 3. Run order: `q1.py`, `q2.py`, `q3.py`, `gen_tables.py`.

## common.py

```
"""Shared helpers for EECE5644 Assignment 1."""
import os
import numpy as np

# output folders used by every script
os.makedirs("out", exist_ok=True)
os.makedirs("figs", exist_ok=True)


def gauss_logpdf(X, m, C):
    """Log of the multivariate Gaussian pdf g(x|m,C), evaluated row-wise on X (N x d).
    Uses a Cholesky factorisation for numerical stability."""
    X = np.atleast_2d(X)
    d = X.shape[1]
    L = np.linalg.cholesky(C)
    Z = np.linalg.solve(L, (X - m).T) # d x N, Z = L^{-1}(x - m)
    maha = np.sum(Z ** 2, axis=0) # (x-m)^T C^{-1} (x-m)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (d * np.log(2 * np.pi) + logdet + maha)


def roc_sweep(score, labels):
    """Sweep a threshold t over all observed score values for the rule D = 1 iff score > t.
    Returns thresholds (starting at -inf, so the curve starts at (1,1)), P(D=1|L=0) and P(D=1|L=1).
    The largest observed score gives the finite end point (0,0)."""
    score = np.asarray(score, float)
    labels = np.asarray(labels).astype(int)
    n1, n0 = labels.sum(), (1 - labels).sum()
    order = np.argsort(score)
    s_sorted = score[order]
    l_sorted = labels[order]
    # number of samples with score <= s_sorted[k] -> those are decided 0 at threshold s_sorted[k]
    c1_le = np.cumsum(l_sorted)
    c0_le = np.cumsum(1 - l_sorted)
    # keep only the last index of each run of tied scores
    last = np.r_[s_sorted[1:] != s_sorted[:-1], True]
    thr = np.r_[-np.inf, s_sorted[last]]
    tpr = np.r_[1.0, (n1 - c1_le[last]) / n1]
    fpr = np.r_[1.0, (n0 - c0_le[last]) / n0]
    return thr, fpr, tpr


def auc(fpr, tpr):
    o = np.argsort(fpr, kind="stable")
    return np.trapezoid(tpr[o], fpr[o])
```

## q1.py

```
"""Question 1: ERM / LRT with true pdf, Naive Bayes, and Fisher LDA on a 2-class 4-D Gaussian problem."""
import json
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy.linalg import eigh
from common import gauss_logpdf, roc_sweep, auc

rng = np.random.default_rng(5644)
N = 10000
P = np.array([0.35, 0.65])
m0 = -0.5 * np.ones(4)
C0 = 0.25 * np.array([[2, -0.5, 0.3, 0],
                      [-0.5, 1, -0.5, 0],
                      [0.3, -0.5, 1, 0],
                      [0, 0, 0, 2]])
m1 = np.ones(4)
C1 = np.array([[1, 0.3, -0.2, 0],
               [0.3, 2, 0.3, 0],
               [-0.2, 0.3, 1, 0],
               [0, 0, 0, 3]])
```

```
for C in (C0, C1):
    assert np.all(np.linalg.eigvalsh(C) > 0)


def sample(n, rng):
    L = (rng.random(n) >= P[0]).astype(int) # P(L=1)=0.65
    X = np.empty((n, 4))
    X[L == 0] = rng.multivariate_normal(m0, C0, (L == 0).sum())
    X[L == 1] = rng.multivariate_normal(m1, C1, (L == 1).sum())
    return X, L


X, L = sample(N, rng)
np.savez("out/q1_data.npz", X=X, L=L) # same data set used for every part
N0, N1 = (L == 0).sum(), (L == 1).sum()
res = {"N0": int(N0), "N1": int(N1)}

gamma_theory = P[0] / P[1] # 0-1 loss: (l10-l00)P0 / ((l01-l11)P1)
res["gamma_theory"] = gamma_theory


def analyse(score, name, theory_thr=None, priors=P):
    """score: decision statistic, D=1 iff score > thr. Returns dict + ROC arrays."""
    thr, fpr, tpr = roc_sweep(score, L)
    perr = fpr * priors[0] + (1 - tpr) * priors[1]
    k = int(np.argmin(perr))
    # any threshold in [thr_k, thr_{k+1}) gives the same decisions; report the midpoint
    t_lo = thr[k]
    t_hi = thr[k + 1] if k + 1 < len(thr) else np.inf
    t_mid = 0.5 * (t_lo + t_hi) if np.isfinite(t_lo) and np.isfinite(t_hi) else t_lo
    out = dict(name=name, k=k, thr_mid=float(t_mid), thr_lo=float(t_lo), thr_hi=float(t_hi),
               perr_min=float(perr[k]), tpr=float(tpr[k]), fpr=float(fpr[k]),
               auc=float(auc(fpr, tpr)), n_thr=len(thr))
    if theory_thr is not None:
        D = score > theory_thr
        out["theory_tpr"] = float(D[L == 1].mean())
        out["theory_fpr"] = float(D[L == 0].mean())
        out["theory_perr"] = float(out["theory_fpr"] * P[0] + (1 - out["theory_tpr"]) * P[1])
    return out, (fpr, tpr, perr, thr)


# ---------- Part A: true pdf ----------
llr_A = gauss_logpdf(X, m1, C1) - gauss_logpdf(X, m0, C0) # ln p(x|1) - ln p(x|0)
rA, curA = analyse(llr_A, "True pdf (ERM)", np.log(gamma_theory))
rA["gamma_emp"] = float(np.exp(rA["thr_mid"]))
rA["gamma_emp_lo"] = float(np.exp(rA["thr_lo"]))
rA["gamma_emp_hi"] = float(np.exp(rA["thr_hi"]))

# ---------- Part B: naive Bayes (diagonal covariances with true variances) ----------
C0d, C1d = np.diag(np.diag(C0)), np.diag(np.diag(C1))
llr_B = gauss_logpdf(X, m1, C1d) - gauss_logpdf(X, m0, C0d)
rB, curB = analyse(llr_B, "Naive Bayes", np.log(gamma_theory))
rB["gamma_emp"] = float(np.exp(rB["thr_mid"]))
rB["gamma_emp_lo"] = float(np.exp(rB["thr_lo"]))
rB["gamma_emp_hi"] = float(np.exp(rB["thr_hi"]))

# ---------- Part C: Fisher LDA from sample estimates ----------
mu0_hat, mu1_hat = X[L == 0].mean(0), X[L == 1].mean(0)
S0_hat = np.cov(X[L == 0].T, bias=True) # sample-average (1/N_j) estimators
S1_hat = np.cov(X[L == 1].T, bias=True)
Sb = np.outer(mu1_hat - mu0_hat, mu1_hat - mu0_hat)
Sw = S0_hat + S1_hat # equal weights, priors ignored (as in class)
evals, evecs = eigh(Sb, Sw) # generalised eigenproblem Sb w = lambda Sw w
w = evecs[:, np.argmax(evals)]
w = w / np.linalg.norm(w)
if w @ mu1_hat < w @ mu0_hat: # orient so that class 1 projects higher
    w = -w
y = X @ w
rC, curC = analyse(y, "Fisher LDA", priors=np.array([N0, N1]) / N) # sample-count estimate
rC["count_check"] = float(np.mean((y > rC["thr_mid"]) != L))
w_closed = np.linalg.solve(Sw, mu1_hat - mu0_hat)
w_closed /= np.linalg.norm(w_closed)
res["lda"] = dict(w=w.tolist(), w_closed=w_closed.tolist(), eig=float(evals.max()),
                  mu0_hat=mu0_hat.tolist(), mu1_hat=mu1_hat.tolist(),
                  S0_hat=S0_hat.tolist(), S1_hat=S1_hat.tolist())

# ---------- reference: theoretical min P(error) via a large independent Monte Carlo ----------
Xbig, Lbig = sample(2_000_000, np.random.default_rng(1))
llr_big = gauss_logpdf(Xbig, m1, C1) - gauss_logpdf(Xbig, m0, C0)
res["perr_bayes_mc"] = float(np.mean((llr_big > np.log(gamma_theory)) != Lbig))
llr_bigB = gauss_logpdf(Xbig, m1, C1d) - gauss_logpdf(Xbig, m0, C0d)
res["perr_nb_mc_theorythr"] = float(np.mean((llr_bigB > np.log(gamma_theory)) != Lbig))
```

```
res["A"], res["B"], res["C"] = rA, rB, rC
json.dump(res, open("out/q1_results.json", "w"), indent=2)

# ---------- figures ----------
def roc_plot(ax, cur, r, title, color, theory=True):
    fpr, tpr, _, _ = cur
    ax.plot(fpr, tpr, color=color, lw=1.6, label=f"ROC (AUC = {r['auc']:.4f})")
    ax.plot(r["fpr"], r["tpr"], "o", ms=10, mfc="none", mec="red", mew=2.2,
            label=f"Empirical min-P(error): P(e) = {r['perr_min']:.4f}")
    if theory and "theory_tpr" in r:
        ax.plot(r["theory_fpr"], r["theory_tpr"], "x", ms=10, color="black", mew=2.2,
                label=f"Theoretical $\\gamma^*=0.538$: P(e) = {r['theory_perr']:.4f}")
    ax.plot([0, 1], [0, 1], ":", color="gray", lw=0.8)
    ax.set_xlabel(r"False positive rate $P(D=1\mid L=0)$")
    ax.set_ylabel(r"True positive rate $P(D=1\mid L=1)$")
    ax.set_title(title)
    ax.legend(loc="lower right", fontsize=8.5)
    ax.grid(alpha=0.3)
    ax.set_xlim(-0.01, 1.01); ax.set_ylim(-0.01, 1.01)


for cur, r, t, c, fn in [(curA, rA, "Part A: ERM with true pdf", "C0", "q1a_roc"),
                         (curB, rB, "Part B: Naive Bayes (diagonal covariances)", "C2", "q1b_roc"),
                         (curC, rC, r"Part C: Fisher LDA, $w^T x$ vs. $\tau$", "C1", "q1c_roc")]:
    fig, ax = plt.subplots(figsize=(6.2, 5.2))
    roc_plot(ax, cur, r, t, c)
    # zoomed inset near the operating point
    ins = ax.inset_axes([0.45, 0.32, 0.5, 0.36])
    ins.plot(cur[0], cur[1], color=c, lw=1.4)
    ins.plot(r["fpr"], r["tpr"], "o", ms=8, mfc="none", mec="red", mew=2)
    if "theory_tpr" in r:
        ins.plot(r["theory_fpr"], r["theory_tpr"], "x", ms=8, color="black", mew=2)
    ins.set_xlim(0, 0.2); ins.set_ylim(0.8, 1.0); ins.grid(alpha=0.3); ins.tick_params(labelsize=7)
    fig.tight_layout(); fig.savefig(f"figs/{fn}.png", dpi=170); plt.close(fig)

# comparison plot
fig, ax = plt.subplots(figsize=(6.2, 5.2))
for cur, r, c in [(curA, rA, "C0"), (curB, rB, "C2"), (curC, rC, "C1")]:
    ax.plot(cur[0], cur[1], color=c, lw=1.5, label=f"{r['name']} (AUC {r['auc']:.4f}, min P(e) {r['perr_min']:.4f})")
    ax.plot(r["fpr"], r["tpr"], "o", ms=8, mfc="none", mec=c, mew=2)
ax.set_xlim(0, 0.3); ax.set_ylim(0.7, 1.0); ax.grid(alpha=0.3)
ax.set_xlabel(r"$P(D=1\mid L=0)$"); ax.set_ylabel(r"$P(D=1\mid L=1)$")
ax.set_title("ROC comparison (zoomed); circles = min-P(error) points")
ax.legend(fontsize=8, loc="lower right")
fig.tight_layout(); fig.savefig("figs/q1_compare.png", dpi=170); plt.close(fig)

# P(error) vs threshold
fig, axs = plt.subplots(1, 3, figsize=(13, 3.6))
for ax, cur, r, xl in [(axs[0], curA, rA, r"$\ln\gamma$"), (axs[1], curB, rB, r"$\ln\gamma$"),
                       (axs[2], curC, rC, r"$\tau$")]:
    _, _, perr, thr = cur
    ok = np.isfinite(thr)
    ax.plot(thr[ok], perr[ok], lw=1.2)
    ax.axvline(r["thr_mid"], color="red", ls="--", lw=1, label=f"empirical min at {r['thr_mid']:.3f}")
    if r is not rC:
        ax.axvline(np.log(gamma_theory), color="k", ls=":", lw=1.2, label=fr"$\ln\gamma^*$ = {np.log(gamma_theory):.3f}")
    lo, hi = np.percentile(thr[ok], [2, 98])
    ax.set_xlim(lo, hi); ax.set_xlabel(xl); ax.set_ylabel("P(error)"); ax.set_title(r["name"])
    ax.legend(fontsize=8); ax.grid(alpha=0.3)
fig.tight_layout(); fig.savefig("figs/q1_perr_vs_thr.png", dpi=160); plt.close(fig)

print(json.dumps({k: v for k, v in res.items() if k != "lda"}, indent=1))
print("w", w, "w_closed", w_closed)
```

## q2.py

```
"""Question 2: 3-class, 3-D Gaussian-mixture problem; MAP (0-1 loss) and ERM with Lambda_10, Lambda_100."""
import json
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.lines import Line2D
from common import gauss_logpdf

rng = np.random.default_rng(2026)
N = 10000
priors = np.array([0.3, 0.3, 0.4])
```

```
# Four component means on a regular tetrahedron with edge length a = 2.5
a = 2.5
means = np.array([[0, 0, 0],
                  [a, 0, 0],
                  [a / 2, a * np.sqrt(3) / 2, 0],
                  [a / 2, a * np.sqrt(3) / 6, a * np.sqrt(2 / 3)]])
covs = np.array([[[1.0, 0.3, -0.2], [0.3, 0.8, 0.1], [-0.2, 0.1, 1.2]],
                 [[0.9, -0.2, 0.1], [-0.2, 1.1, 0.3], [0.1, 0.3, 0.8]],
                 [[1.2, 0.2, 0.0], [0.2, 0.9, -0.3], [0.0, -0.3, 1.0]],
                 [[0.8, 0.0, 0.2], [0.0, 1.0, 0.0], [0.2, 0.0, 1.1]]])
for C in covs:
    assert np.all(np.linalg.eigvalsh(C) > 0)
# class 1 <- component 0, class 2 <- component 1, class 3 <- 0.5*comp 2 + 0.5*comp 3
avg_std = np.mean([np.sqrt(np.trace(C) / 3) for C in covs])
dists = [np.linalg.norm(means[i] - means[j]) for i in range(4) for j in range(i + 1, 4)]

# ---- 1. generate data ----
L = rng.choice([1, 2, 3], size=N, p=priors)
comp = np.where(L == 1, 0, np.where(L == 2, 1, 2 + (rng.random(N) < 0.5)))
X = np.empty((N, 3))
for c in range(4):
    idx = comp == c
    X[idx] = rng.multivariate_normal(means[c], covs[c], idx.sum())
np.savez("out/q2_data.npz", X=X, L=L, comp=comp)


def class_loglik(X):
    """N x 3 matrix of ln p(x|L=j)."""
    l1 = gauss_logpdf(X, means[0], covs[0])
    l2 = gauss_logpdf(X, means[1], covs[1])
    l3 = np.logaddexp(np.log(0.5) + gauss_logpdf(X, means[2], covs[2]),
                      np.log(0.5) + gauss_logpdf(X, means[3], covs[3]))
    return np.c_[l1, l2, l3]


ll = class_loglik(X)
lj = ll + np.log(priors) # ln p(x|j)P(j)
post = np.exp(lj - np.logaddexp.reduce(lj, axis=1, keepdims=True)) # P(L=j|x)

Lambda = {"0-1": 1 - np.eye(3),
          "10": np.array([[0, 1, 10], [1, 0, 10], [1, 1, 0]], float),
          "100": np.array([[0, 1, 100], [1, 0, 100], [1, 1, 0]], float)}

res = {"means": means.tolist(), "covs": covs.tolist(), "avg_std": avg_std,
       "pairwise_dist": dists, "class_counts": [int((L == j).sum()) for j in (1, 2, 3)]}

for key, Lam in Lambda.items():
    R = post @ Lam.T # R[n,i] = sum_j Lam[i,j] P(j|x_n) (conditional risk)
    D = np.argmin(R, axis=1) + 1
    conf = np.zeros((3, 3))
    counts = np.zeros((3, 3), int)
    for i in (1, 2, 3):
        for j in (1, 2, 3):
            counts[i - 1, j - 1] = np.sum((D == i) & (L == j))
            conf[i - 1, j - 1] = counts[i - 1, j - 1] / np.sum(L == j)
    risk = float(np.mean(Lam[D - 1, L - 1])) # sample average of incurred loss
    perr = float(np.mean(D != L))
    res[key] = dict(conf=conf.tolist(), counts=counts.tolist(), risk=risk, perr=perr,
                    dec_counts=[int((D == i).sum()) for i in (1, 2, 3)],
                    exp_risk_cond=float(np.mean(R.min(axis=1))))
    np.save(f"out/q2_D_{key}.npy", D)

    # ---- 3-D scatter plot ----
    fig = plt.figure(figsize=(12, 6.2))
    markers = {1: ".", 2: "^", 3: "s"}
    sizes = {1: 30, 2: 10, 3: 8}
    ok = D == L
    for p, (elev, azim) in enumerate([(22, -60), (12, 30)]):
        ax = fig.add_subplot(1, 2, p + 1, projection="3d")
        for j in (1, 2, 3):
            for good, col in ((True, "green"), (False, "red")):
                idx = (L == j) & (ok == good)
                ax.scatter(*X[idx].T, marker=markers[j], s=sizes[j],
                           c=col, alpha=0.35 if good else 0.8, linewidths=0.3)
        ax.view_init(elev, azim)
        ax.set_box_aspect(None, zoom=1.0)
        ax.set_xlabel("$x_1$"); ax.set_ylabel("$x_2$"); ax.set_zlabel("$x_3$")
    handles = [Line2D([], [], marker=markers[j], ls="", color="gray", ms=7, label=f"L = {j}") for j in (1, 2, 3)]
    handles += [Line2D([], [], marker="o", ls="", color="green", label="correct"),
                Line2D([], [], marker="o", ls="", color="red", label="incorrect")]
    name = "MAP (0-1 loss)" if key == "0-1" else fr"ERM with $\Lambda_{{{key}}}$"
```

```
    fig.suptitle(f"{name}: P(error) = {perr:.4f}, estimated min. expected risk = {risk:.4f}")
    fig.legend(handles=handles, loc="lower center", ncol=5, fontsize=9)
    fig.subplots_adjust(left=0.02, right=0.98, top=0.95, bottom=0.07, wspace=0.08)
    fig.savefig(f"figs/q2_scatter_{key.replace('-', '')}.png", dpi=150); plt.close(fig)

# ---- confusion-matrix heat maps side by side ----
fig, axs = plt.subplots(1, 3, figsize=(12, 3.8))
for ax, key in zip(axs, Lambda):
    M = np.array(res[key]["conf"])
    ax.imshow(M, cmap="Blues", vmin=0, vmax=1)
    for i in range(3):
        for j in range(3):
            ax.text(j, i, f"{M[i, j]:.3f}", ha="center", va="center",
                    color="white" if M[i, j] > 0.6 else "black", fontsize=10)
    ax.set_xticks(range(3), ["L=1", "L=2", "L=3"]); ax.set_yticks(range(3), ["D=1", "D=2", "D=3"])
    t = "0-1 loss (MAP)" if key == "0-1" else fr"$\Lambda_{{{key}}}$"
    ax.set_title(f"{t}\nrisk {res[key]['risk']:.4f}, P(e) {res[key]['perr']:.4f}", fontsize=10)
fig.suptitle(r"Confusion matrices $P(D=i\mid L=j)$ (columns sum to 1)")
fig.tight_layout(); fig.savefig("figs/q2_confusions.png", dpi=150); plt.close(fig)

# ---- risk/error trade-off as the class-3 loss weight grows ----
ws = np.r_[1, 2, 3, 5, 7, 10, 15, 20, 30, 50, 70, 100, 200, 500]
rk, pe, p3 = [], [], []
for wgt in ws:
    Lam = np.array([[0, 1, wgt], [1, 0, wgt], [1, 1, 0]], float)
    D = np.argmin(post @ Lam.T, axis=1) + 1
    rk.append(np.mean(Lam[D - 1, L - 1])); pe.append(np.mean(D != L)); p3.append(np.mean(D == 3))
res["sweep"] = dict(w=ws.tolist(), risk=rk, perr=pe, frac_D3=p3)
fig, ax = plt.subplots(figsize=(6.4, 4))
ax.semilogx(ws, pe, "o-", label="P(error)")
ax.semilogx(ws, p3, "s-", label="fraction decided D=3")
ax.semilogx(ws, rk, "^-", label="estimated expected risk")
ax.set_xlabel("loss for misclassifying a class-3 sample (w)"); ax.grid(alpha=0.3, which="both")
ax.legend(); ax.set_title("Effect of the class-3 loss weight")
fig.tight_layout(); fig.savefig("figs/q2_sweep.png", dpi=150); plt.close(fig)

json.dump(res, open("out/q2_results.json", "w"), indent=2)
for key in Lambda:
    print(key, res[key]["risk"], res[key]["perr"], res[key]["dec_counts"])
    print(np.round(np.array(res[key]["conf"]), 4))
print("avg std", avg_std, "dists", np.round(dists, 3), res["class_counts"])
```

## q3.py

```
"""Question 3: Gaussian class-conditional (QDA-type) minimum-P(error) classifiers on real data."""
import json
import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy.stats import chi2
from common import gauss_logpdf

DATA = os.path.join(os.path.dirname(os.path.abspath(__file__)), "data")


# ---------------------------------------------------------------------- data
def load_wine():
    df = pd.read_csv(f"{DATA}/winequality-white.csv", sep=";")
    X = df.drop(columns="quality").to_numpy(float)
    y = df["quality"].to_numpy(int)
    return X, y, list(df.columns[:-1])


def load_har():
    Xtr = np.loadtxt(f"{DATA}/har/X_train.txt"); Xte = np.loadtxt(f"{DATA}/har/X_test.txt")
    ytr = np.loadtxt(f"{DATA}/har/y_train.txt", int); yte = np.loadtxt(f"{DATA}/har/y_test.txt", int)
    names = [l.split()[1] for l in open(f"{DATA}/har/activity_labels.txt")]
    return np.vstack([Xtr, Xte]), np.r_[ytr, yte], names, len(ytr)


# ---------------------------------------------------------------------- model
def fit(X, y, alpha):
    """Per-class priors (counts), sample-average means and covariances, plus
    lambda_j = alpha * trace(C_j) / rank(C_j) added to the diagonal."""
    model = {}
    for c in np.unique(y):
```

```
        Xc = X[y == c]
        m = Xc.mean(0)
        C = (Xc - m).T @ (Xc - m) / len(Xc) # 1/N_c sample-average estimator
        r = np.linalg.matrix_rank(C)
        lam = alpha * np.trace(C) / r
        model[c] = dict(prior=len(Xc) / len(X), m=m, C=C + lam * np.eye(X.shape[1]), lam=lam,
                        rank=int(r), n=len(Xc), cond_raw=float(np.linalg.cond(C)),
                        cond_reg=float(np.linalg.cond(C + lam * np.eye(X.shape[1]))))
    return model


def predict(model, X):
    cls = np.array(sorted(model))
    S = np.column_stack([gauss_logpdf(X, model[c]["m"], model[c]["C"]) + np.log(model[c]["prior"]) for c in cls])
    return cls[np.argmax(S, 1)]


def confusion(D, y, cls):
    """M[i,j] = P(D=cls_i | L=cls_j), counts as well."""
    K = len(cls)
    cnt = np.zeros((K, K), int)
    for a, ci in enumerate(cls):
        for b, cj in enumerate(cls):
            cnt[a, b] = np.sum((D == ci) & (y == cj))
    return cnt / cnt.sum(0, keepdims=True), cnt


def stratified_folds(y, k, rng):
    f = np.empty(len(y), int)
    for c in np.unique(y):
        idx = rng.permutation(np.where(y == c)[0])
        f[idx] = np.arange(len(idx)) % k
    return f


def heat(ax, M, cnt, ticks, title):
    ax.imshow(M, cmap="Blues", vmin=0, vmax=1)
    for i in range(M.shape[0]):
        for j in range(M.shape[1]):
            if cnt[:, j].sum() == 0:
                continue
            ax.text(j, i, f"{M[i, j]:.2f}\n({cnt[i, j]})", ha="center", va="center", fontsize=7,
                    color="white" if M[i, j] > 0.6 else "black")
    ax.set_xticks(range(len(ticks)), [f"L={t}" for t in ticks], fontsize=8, rotation=0)
    ax.set_yticks(range(len(ticks)), [f"D={t}" for t in ticks], fontsize=8)
    ax.set_title(title, fontsize=10)


def pca(X, k):
    Xc = X - X.mean(0)
    U, s, Vt = np.linalg.svd(Xc, full_matrices=False)
    return Xc @ Vt[:k].T, s ** 2 / np.sum(s ** 2)


res = {}
rng = np.random.default_rng(0)
alphas = [1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 3e-2, 1e-1, 3e-1]

# ================================================================ WINE
Xw_raw, yw, wnames = load_wine()
mu_w, sd_w = Xw_raw.mean(0), Xw_raw.std(0)
Xw = (Xw_raw - mu_w) / sd_w # z-score (see report for why)
cls_w = np.unique(yw)
res["wine_counts"] = {int(c): int((yw == c).sum()) for c in cls_w}

folds = stratified_folds(yw, 5, rng)
sweep_w = []
for a in alphas:
    m = fit(Xw, yw, a)
    tr_err = float(np.mean(predict(m, Xw) != yw))
    cv_err = []
    for f in range(5):
        mf = fit(Xw[folds != f], yw[folds != f], a)
        cv_err.append(np.mean(predict(mf, Xw[folds == f]) != yw[folds == f]))
    sweep_w.append(dict(alpha=a, train=tr_err, cv=float(np.mean(cv_err))))
res["wine_sweep"] = sweep_w
alpha_w = min(sweep_w, key=lambda r: r["cv"])["alpha"]
mw = fit(Xw, yw, alpha_w)
Dw = predict(mw, Xw)
Mw, Cw = confusion(Dw, yw, cls_w)
res["wine"] = dict(alpha=alpha_w, perr=float(np.mean(Dw != yw)), conf=Mw.tolist(), counts=Cw.tolist(),
                   classes=cls_w.tolist(),
```

```
                per_class={int(c): dict(n=mw[c]["n"], prior=mw[c]["prior"], rank=mw[c]["rank"],
                                        lam=mw[c]["lam"], cond_raw=mw[c]["cond_raw"],
                                        cond_reg=mw[c]["cond_reg"]) for c in cls_w},
                majority_err=float(1 - max(np.mean(yw == c) for c in cls_w)))
# same classifier on raw (unstandardised) features, for comparison
mw_raw = fit(Xw_raw, yw, alpha_w)
res["wine"]["perr_raw_features"] = float(np.mean(predict(mw_raw, Xw_raw) != yw))
# within-one-grade accuracy
res["wine"]["within1"] = float(np.mean(np.abs(Dw - yw) <= 1))

# ================================================================ HAR
Xh, yh, hnames, ntr = load_har()
mu_h, sd_h = Xh.mean(0), Xh.std(0)
Xhs = (Xh - mu_h) / sd_h
cls_h = np.unique(yh)
res["har_counts"] = {int(c): int((yh == c).sum()) for c in cls_h}
sweep_h = []
for a in alphas:
    m = fit(Xhs, yh, a)
    tr_err = float(np.mean(predict(m, Xhs) != yh))
    # held-out check on the dataset's own subject-disjoint train/test split
    mt = fit(Xhs[:ntr], yh[:ntr], a)
    te_err = float(np.mean(predict(mt, Xhs[ntr:]) != yh[ntr:]))
    sweep_h.append(dict(alpha=a, train=tr_err, test=te_err))
    print("HAR", a, tr_err, te_err, flush=True)
res["har_sweep"] = sweep_h
alpha_h = min(sweep_h, key=lambda r: r["test"])["alpha"]
mh = fit(Xhs, yh, alpha_h)
Dh = predict(mh, Xhs)
Mh, Ch = confusion(Dh, yh, cls_h)
res["har"] = dict(alpha=alpha_h, perr=float(np.mean(Dh != yh)), conf=Mh.tolist(), counts=Ch.tolist(),
                  names=hnames,
                  per_class={int(c): dict(n=mh[c]["n"], prior=mh[c]["prior"], rank=mh[c]["rank"],
                                          lam=mh[c]["lam"], cond_raw=mh[c]["cond_raw"],
                                          cond_reg=mh[c]["cond_reg"]) for c in cls_h})
json.dump(res, open("out/q3_results.json", "w"), indent=2, default=float)

# ================================================================ FIGURES
# confusion matrices
fig, ax = plt.subplots(figsize=(6.4, 5.4))
heat(ax, Mw, Cw, cls_w, fr"Wine (white): $P(D=i\mid L=j)$, P(error) = {res['wine']['perr']:.4f}")
fig.tight_layout(); fig.savefig("figs/q3_wine_conf.png", dpi=160); plt.close(fig)
fig, ax = plt.subplots(figsize=(6.4, 5.4))
heat(ax, Mh, Ch, cls_h, fr"HAR: $P(D=i\mid L=j)$, P(error) = {res['har']['perr']:.4f}")
fig.tight_layout(); fig.savefig("figs/q3_har_conf.png", dpi=160); plt.close(fig)

# regularisation sweep
fig, axs = plt.subplots(1, 2, figsize=(11, 3.8))
axs[0].semilogx(alphas, [r["train"] for r in sweep_w], "o-", label="all-sample (training) error")
axs[0].semilogx(alphas, [r["cv"] for r in sweep_w], "s-", label="5-fold CV error")
axs[0].axvline(alpha_w, color="k", ls=":"); axs[0].set_title("Wine"); axs[0].set_xlabel(r"$\alpha$")
axs[1].semilogx(alphas, [r["train"] for r in sweep_h], "o-", label="all-sample (training) error")
axs[1].semilogx(alphas, [r["test"] for r in sweep_h], "s-", label="held-out error (official split)")
axs[1].axvline(alpha_h, color="k", ls=":"); axs[1].set_title("HAR"); axs[1].set_xlabel(r"$\alpha$")
for a in axs:
    a.set_ylabel("P(error)"); a.grid(alpha=0.3, which="both"); a.legend(fontsize=8)
fig.tight_layout(); fig.savefig("figs/q3_alpha_sweep.png", dpi=160); plt.close(fig)

# PCA projections
cmap = plt.get_cmap("tab10")
for nm, X, y, cls, lab in [("wine", Xw, yw, cls_w, lambda c: f"q={c}"),
                           ("har", Xhs, yh, cls_h, lambda c: hnames[c - 1])]:
    Z, evr = pca(X, 3)
    fig = plt.figure(figsize=(13, 4.6))
    ax1 = fig.add_subplot(1, 3, 1); ax2 = fig.add_subplot(1, 3, 2); ax3 = fig.add_subplot(1, 3, 3, projection="3d")
    for k, c in enumerate(cls):
        i = y == c
        kw = dict(s=4, alpha=0.45, color=cmap(k), label=lab(c))
        ax1.scatter(Z[i, 0], Z[i, 1], **kw); ax2.scatter(Z[i, 0], Z[i, 2], **kw)
        ax3.scatter(Z[i, 0], Z[i, 1], Z[i, 2], **kw)
    ax1.set_xlabel(f"PC1 ({evr[0]:.1%})"); ax1.set_ylabel(f"PC2 ({evr[1]:.1%})")
    ax2.set_xlabel("PC1"); ax2.set_ylabel(f"PC3 ({evr[2]:.1%})")
    ax3.set_xlabel("PC1"); ax3.set_ylabel("PC2"); ax3.set_zlabel("PC3")
    ax1.legend(markerscale=4, fontsize=7, loc="best")
    fig.suptitle(f"{'White wine' if nm == 'wine' else 'HAR'}: first three principal components (standardised features)")
    fig.tight_layout(); fig.savefig(f"figs/q3_{nm}_pca.png", dpi=150); plt.close(fig)

# wine: feature-subset scatter + marginal histograms
fig, axs = plt.subplots(1, 3, figsize=(13, 4))
pairs = [("alcohol", "density"), ("alcohol", "volatile acidity"), ("residual sugar", "chlorides")]
for ax, (fx, fy) in zip(axs, pairs):
    ix, iy = wnames.index(fx), wnames.index(fy)
```

```
    for k, c in enumerate(cls_w):
        i = yw == c
        ax.scatter(Xw_raw[i, ix], Xw_raw[i, iy], s=4, alpha=0.5, color=cmap(k), label=f"q={c}")
    ax.set_xlabel(fx); ax.set_ylabel(fy)
axs[0].legend(markerscale=4, fontsize=7)
fig.suptitle("White wine: raw feature pairs")
fig.tight_layout(); fig.savefig("figs/q3_wine_pairs.png", dpi=150); plt.close(fig)

fig, axs = plt.subplots(2, 6, figsize=(15, 5))
for k, ax in enumerate(axs.ravel()):
    if k >= len(wnames):
        ax.axis("off"); continue
    ax.hist(Xw_raw[yw == 6, k], bins=50, density=True, color="C0", alpha=0.7)
    ax.set_title(wnames[k], fontsize=9); ax.tick_params(labelsize=7)
fig.suptitle("White wine, quality-6 class: marginal histograms (a Gaussian model implies symmetric, bell-shaped marginals)")
fig.tight_layout(); fig.savefig("figs/q3_wine_hist.png", dpi=140); plt.close(fig)

# HAR: a few raw features
fig, axs = plt.subplots(1, 2, figsize=(11, 4.2))
for ax, (fx, fy) in zip(axs, [(0, 1), (40, 52)]): # 0-based column indices into the 561 features
    for k, c in enumerate(cls_h):
        i = yh == c
        ax.scatter(Xh[i, fx], Xh[i, fy], s=3, alpha=0.4, color=cmap(k), label=hnames[c - 1])
    ax.set_xlabel(f"feature {fx + 1}"); ax.set_ylabel(f"feature {fy + 1}")
axs[0].legend(markerscale=4, fontsize=7)
fig.suptitle("HAR: two raw feature pairs")
fig.tight_layout(); fig.savefig("figs/q3_har_pairs.png", dpi=150); plt.close(fig)

# Gaussianity check: squared Mahalanobis distances vs chi^2_d quantiles (Q-Q)
fig, axs = plt.subplots(1, 2, figsize=(11, 4.3))
for ax, (nm, X, y, mdl, cs) in zip(axs, [("Wine", Xw, yw, mw, [5, 6, 7]), ("HAR", Xhs, yh, mh, [1, 4, 6])]):
    d = X.shape[1]
    for c in cs:
        Xc = X[y == c]
        m = Xc.mean(0); C = mdl[c]["C"]
        Lc = np.linalg.cholesky(C)
        r2 = np.sort(np.sum(np.linalg.solve(Lc, (Xc - m).T) ** 2, 0))
        q = chi2.ppf((np.arange(1, len(r2) + 1) - 0.5) / len(r2), d)
        lab = f"q={c}" if nm == "Wine" else hnames[c - 1]
        ax.plot(q, r2, ".", ms=2, label=lab)
    lim = ax.get_xlim()
    ax.plot(lim, lim, "k--", lw=1, label="y = x (Gaussian)")
    ax.set_xlabel(fr"$\chi^2_{{{d}}}$ quantiles"); ax.set_ylabel("sorted squared Mahalanobis distance")
    ax.set_title(f"{nm}: Q-Q check of the Gaussian model"); ax.legend(markerscale=5, fontsize=8)
    if nm == "Wine":
        ax.set_ylim(0, 150)
    ax.grid(alpha=0.3)
fig.tight_layout(); fig.savefig("figs/q3_qq.png", dpi=150); plt.close(fig)

print(json.dumps({k: res[k] for k in ("wine_sweep", "har_sweep")}, indent=1))
print("wine", alpha_w, res["wine"]["perr"], res["wine"]["perr_raw_features"], res["wine"]["within1"], res["wine"]["majority_err"])
print(np.round(Mw, 3)); print(Cw)
print("har", alpha_h, res["har"]["perr"]); print(np.round(Mh, 4)); print(Ch)
for c in cls_w: print(c, res["wine"]["per_class"][int(c)])
for c in cls_h: print(c, res["har"]["per_class"][int(c)])
```

## gen_tables.py

```
"""Write LaTeX table fragments from the saved JSON results (avoids hand-copying numbers)."""
import json
import numpy as np

q1 = json.load(open("out/q1_results.json"))
q2 = json.load(open("out/q2_results.json"))
q3 = json.load(open("out/q3_results.json"))


def conf_table(M, rows, cols, fmt="{:.4f}", counts=None):
    M = np.array(M)
    s = "\\begin{tabular}{l" + "r" * len(cols) + "}\n\\toprule\n"
    s += " & " + " & ".join(cols) + " \\\\\n\\midrule\n"
    for i, r in enumerate(rows):
        cells = []
        for j in range(len(cols)):
            v = fmt.format(M[i, j])
            if i == j:
                v = "\\textbf{" + v + "}"
            if counts is not None:
                v += f" ({counts[i][j]})"
```

```
            cells.append(v)
        s += r + " & " + " & ".join(cells) + " \\\\\n"
    return s + "\\bottomrule\n\\end{tabular}\n"


# ---- Q1 summary
A, B, C = q1["A"], q1["B"], q1["C"]
s = "\\begin{tabular}{lccccc}\n\\toprule\n"
s += "Classifier & threshold at min $P(e)$ & $P(D{=}1|L{=}1)$ & $P(D{=}1|L{=}0)$ & $\\hat P(\\text{error})$ & AUC \\\\\n\\midrule\
    n"
s += f"A: true pdf & $\\hat\\gamma = {A['gamma_emp']:.3f}$ ($\\ln\\hat\\gamma={A['thr_mid']:.3f}$) & {A['tpr']:.4f} & {A['fpr']:.4
    f} & {A['perr_min']:.4f} & {A['auc']:.4f} \\\\\n"
s += f"\\quad at theory $\\gamma^*={q1['gamma_theory']:.4f}$ & $\\ln\\gamma^*={np.log(q1['gamma_theory']):.3f}$ & {A['theory_tpr
    ']:.4f} & {A['theory_fpr']:.4f} & {A['theory_perr']:.4f} & \\\\\n"
s += f"B: naive Bayes & $\\hat\\gamma = {B['gamma_emp']:.3f}$ ($\\ln\\hat\\gamma={B['thr_mid']:.3f}$) & {B['tpr']:.4f} & {B['fpr
    ']:.4f} & {B['perr_min']:.4f} & {B['auc']:.4f} \\\\\n"
s += f"\\quad at $\\gamma^*={q1['gamma_theory']:.4f}$ & $\\ln\\gamma^*={np.log(q1['gamma_theory']):.3f}$ & {B['theory_tpr']:.4f} &
     {B['theory_fpr']:.4f} & {B['theory_perr']:.4f} & \\\\\n"
s += f"C: Fisher LDA & $\\hat\\tau = {C['thr_mid']:.4f}$ & {C['tpr']:.4f} & {C['fpr']:.4f} & {C['perr_min']:.4f} & {C['auc']:.4f}
    \\\\\n"
s += "\\bottomrule\n\\end{tabular}\n"
open("out/t_q1.tex", "w").write(s)

# ---- Q2 confusion matrices
for key in ("0-1", "10", "100"):
    t = conf_table(q2[key]["conf"], ["$D=1$", "$D=2$", "$D=3$"], ["$L=1$", "$L=2$", "$L=3$"],
                   counts=q2[key]["counts"])
    open(f"out/t_q2_{key.replace('-', '')}.tex", "w").write(t)
s = "\\begin{tabular}{lccccc}\n\\toprule\nLoss & $\\hat{R}_{\\min}=\\frac1N\\sum_n\\lambda_{D_nL_n}$ & $\\frac1N\\sum_n \\min_i R(
    D{=}i|x_n)$ & $\\hat P(\\text{error})$ & \\#$\\{D=1,2,3\\}$ \\\\\n\\midrule\n"
for key, nm in (("0-1", "0--1"), ("10", "$\\Lambda_{10}$"), ("100", "$\\Lambda_{100}$")):
    r = q2[key]
    s += f"{nm} & {r['risk']:.4f} & {r['exp_risk_cond']:.4f} & {r['perr']:.4f} & {r['dec_counts'][0]}, {r['dec_counts'][1]}, {r['
        dec_counts'][2]} \\\\\n"
s += "\\bottomrule\n\\end{tabular}\n"
open("out/t_q2_sum.tex", "w").write(s)

# ---- Q3
w = q3["wine"]
cls = w["classes"]
open("out/t_q3_wine.tex", "w").write(conf_table(w["conf"], [f"$D={c}$" for c in cls], [f"$L={c}$" for c in cls], "{:.3f}"))
h = q3["har"]
short = ["Walk", "Up", "Down", "Sit", "Stand", "Lay"]
open("out/t_q3_har.tex", "w").write(conf_table(h["conf"], [f"$D$={n}" for n in short], [f"$L$={n}" for n in short], "{:.4f}"))

s = "\\begin{tabular}{lrrrrrr}\n\\toprule\nClass & $N_j$ & $\\hat P(L{=}j)$ & rank$(\\hat C_j)$ & $\\lambda_j$ & cond$(\\hat C_j)$
    & cond$(\\hat C_j+\\lambda_j I)$ \\\\\n\\midrule\n"
for c in cls:
    p = w["per_class"][str(c)]
    s += f"{c} & {p['n']} & {p['prior']:.4f} & {p['rank']} & {p['lam']:.2e} & {p['cond_raw']:.2e} & {p['cond_reg']:.2e} \\\\\n"
s += "\\bottomrule\n\\end{tabular}\n"
open("out/t_q3_wine_cls.tex", "w").write(s)
s = "\\begin{tabular}{lrrrrrr}\n\\toprule\nClass & $N_j$ & $\\hat P(L{=}j)$ & rank$(\\hat C_j)$ & $\\lambda_j$ & cond$(\\hat C_j)$
    & cond$(\\hat C_j+\\lambda_j I)$ \\\\\n\\midrule\n"
for i, c in enumerate(sorted(int(k) for k in h["per_class"])):
    p = h["per_class"][str(c)]
    s += f"{c} {short[i]} & {p['n']} & {p['prior']:.4f} & {p['rank']} & {p['lam']:.2e} & {p['cond_raw']:.1e} & {p['cond_reg']:.2e}
        \\\\\n"
s += "\\bottomrule\n\\end{tabular}\n"
open("out/t_q3_har_cls.tex", "w").write(s)

s = "\\begin{tabular}{lcccc}\n\\toprule\n$\\alpha$ & Wine: all-sample & Wine: 5-fold CV & HAR: all-sample & HAR: held-out (
    official split) \\\\\n\\midrule\n"
for a, b in zip(q3["wine_sweep"], q3["har_sweep"]):
    s += f"{a['alpha']:g} & {a['train']:.4f} & {a['cv']:.4f} & {b['train']:.4f} & {b['test']:.4f} \\\\\n"
s += "\\bottomrule\n\\end{tabular}\n"
open("out/t_q3_sweep.tex", "w").write(s)
print("ok")
```